PHYS4071

Big Bang Cosmology & Inflation

Course Notes

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Introduction

About These Notes

These notes accompany the PHYS4071 Big Bang Cosmology & Inflation course in Hong Kong University of Science and Technology (HKUST). They are intended as a self-contained supplement to the lectures, providing both conceptual grounding and mathematical rigour at a level appropriate for readers with a background in special relativity, basic linear algebra, and basic calculus.

The material covers different cosmological topics, including but not limited to the Friedmann equations, and the energy budget of the ΛCDM model. Each topic is structured to serve different learning objectives—from building physical intuition to solving examination-style problems and exploring advanced derivations.

These notes do not replace lecture slides or the recommended textbooks. They are best used alongside them as a structured study companion.

If you find any errors or have suggestions for improvement, please feel free to contact me at `yhip@connect.ust.hk`.

Document Conventions

Block Components

Each topic is presented using a consistent **5-component block structure** designed to support multiple learning styles:

PHYSICS INTUITION

Purpose: Builds conceptual understanding through plain language explanations.
This section provides the *motivation* behind the physics concept, explores its *historical development*, and offers *intuitive explanations* that help you grasp what the theory is really about before diving into equations.

Use this when: You're first encountering a new concept or need to refresh the "big picture" before tackling mathematics.

Theory

Purpose: Provides rigorous mathematical treatment of the concept. Here you'll find detailed *derivations*, *fundamental assumptions*, key *equations*, and connections between *mathematics and physical intuition*. This is the technical foundation required for exams and deeper problem-solving.

Use this when: You need to understand how to apply formulas, prepare for exams, or work through problems rigorously.

Question Examples

Purpose: Demonstrates practical application through worked problems.
Each question example mimics exam-style problems and includes complete *solution walkthroughs*. These help you bridge the gap between theory and problem-solving, and expose common pitfalls.

Use this when: Practicing problem-solving, preparing for assessments, or checking if you can apply the formulas correctly.

Foundation

Purpose: Supplies background knowledge and detailed proof steps that underpin the main content.
This section presents *prerequisite mathematics or physics*, *step-by-step derivations* that are too detailed to include in the Theory block, and *intermediate results* needed for a thorough understanding. Reading this is optional for the exam but essential for genuine mastery.

Use this when: A derivation in Theory feels too abrupt, you want to verify an intermediate step, or you need to solidify prerequisite concepts.

Extra

Purpose: Provides supplementary material that extends *beyond* the course syllabus.
This section offers *interesting asides*, *historical context*, *connections to current research*, and citations to deeper references. The content here is intentionally out-of-scope for the course examination but enriches the broader picture of each topic.

Use this when: You're curious to explore beyond the syllabus, or when a topic connects to your personal research interests.

Suggested Reading Paths:

1. **Exam Preparation:** *Physics Intuition* → *Theory* → *Question Examples*
2. **Deep Learning:** *Physics Intuition* → *Foundation* → *Theory* → *Question Examples* → *Extra*
3. **Quick Review:** *Theory* → *Question Examples*

Special Annotations

In addition to the five block components above, two inline annotations appear throughout the text to flag specific types of information:

IMPORTANT NOTE

Purpose: Alerts you to a result, condition, or sign convention that is easy to get wrong or is frequently tested.
An Important Note signals a *common misconception*, a *critical sign*, a *non-obvious condition* (e.g. a constraint that must hold for an equation to be valid), or an *exam trap* worth memorising explicitly.

Treat these as mandatory reading regardless of which reading path you follow.

REMARK

Purpose: Highlights a useful observation, connection, or naming convention.
A Remark draws attention to a *conceptual link* between results, an *elegant reformulation*, a *naming convention* (e.g. why a quantity carries its particular name), or a *minor but clarifying point* that does not rise to the level of an Important Note.

Use this when: Looking for a deeper understanding of why an equation takes the form it does, or for useful mnemonics.

Mathematical Conventions

The following conventions are used throughout these notes:

Metric signature. We adopt the *mostly-plus* (or "$(-,+,+,+)$") signature:

$$g_{\mu\nu} = \mathrm{diag}(-1,\ +1,\ +1,\ +1).$$

Important Note

Many general-relativity textbooks use the opposite, *mostly-minus* $(+,-,-,-)$ signature. The two conventions differ by an overall sign in the Ricci scalar and stress-energy trace. Always check the signature convention when comparing formulae across sources.

Natural units. In sections on particle physics or early-universe thermodynamics, we may set $c = \hbar = k_B = 1$. Quantities are then expressed in powers of eV (or GeV). Explicit c and $\hbar$ factors are restored when numerical values are quoted in SI units.

Key symbols.

Symbol	Meaning	Typical units
H_0	Hubble constant today	$\mathrm{km\,s^{-1}\,Mpc^{-1}}$
h	Dimensionless Hubble: $H_0 = 100h$	—
Ω_i	Density parameter for species i	—
$a(t)$	Scale factor ($a_0 = 1$ today)	—
z	Cosmological redshift	—
ρ_{crit}	Critical density	$\mathrm{kg\,m^{-3}}$

Remark

The convention $a_0 \equiv a(t_0) = 1$ is standard and simplifies many formulae, e.g. $1 + z = 1/a(t_e)$. Some older texts normalise a differently; always check the definition when reading the literature.

CHAPTER 1

Fundamentals (lecture 1-2)

Before we can start to build up our universe model, we need to first clarify some basic concepts and notations first for convenience.

1.1 Natural Units

PHYSICS INTUITION

In SI units, we treat time, length, mass, and energy as distinct physical quantities with different dimensions and units.

To make calculations more convenient, we introduce natural units to connect these quantities via fundamental constants, so that we can express everything in terms of a single unit and same type of dimension (i.e. energy).

THEORY

In natural units, we set $c = \hbar = k_B = 1$.
When $c = 1$, mass and energy are measured in the same units and dimension in terms of E^1 by $E = mc^2 = m$.

Question Example 1.1: Dimension of Gravitational Constant

What is the dimension of the gravitational constant G in natural units?

Solution. We know that $F = G\frac{m_1 m_2}{r^2}$ and $F = ma$.
We get dimension of a is E^1 by $a = v/t = x/t^2 = \frac{E^{-1}}{E^{-1} \cdot E^{-1}} = E^1$.
Then we get dimension of F is E^2 by $F = ma = E^1 \cdot E^1 = E^2$.
Finally, we get dimension of G is E^{-2} by $G = \frac{Fr^2}{m_1 m_2} = E^2 \cdot E^{-2}/(E^1 \cdot E^1) = E^{-2}$.

In general, we define the dimension of a quantity directly in terms of n if its dimension is E^n for simplicity. Below are some useful conversion factors between SI and natural units:

Quantity	Related equation	SI dimension	Natural unit
Energy	$E = mc^2$	$\mathrm{kg\,m^2\,s^{-2}}$	1
Mass	$m = E/c^2$	kg	1
Frequency	$\nu = E/h$	Hz	1
Time	$t = 1/\nu$	s	-1
Space	$x = ct$	m	-1
Temperature	$T = E/k_B$	K	1
Velocity	$v = x/t$	$\mathrm{m\,s^{-1}}$	0
Momentum	$p = E/c$	$\mathrm{kg\,m\,s^{-1}}$	1
Angular Velocity	$\omega = rv$	$\mathrm{rad\,s^{-1}}$	-1
Angular Momentum	$L = \frac{1}{2}mv\omega$	$\mathrm{kg\,m^2\,s^{-1}}$	0
Luminosity	$\mathcal{L} = E/t$	W	2
Force	$F = p/t$	N	2
Pressure	$P = F/A$	Pa	2
Mass Density	$\rho = m/V$	$\mathrm{kg\,m^{-3}}$	4
Energy Density	$\rho = E/V$	$\mathrm{J\,m^{-3}}$	4
Gravitational Constant	$G = Fr^2/m_1m_2$	$\mathrm{m^3\,kg^{-1}\,s^{-2}}$	-2
Redshift	$z = \lambda_0/\lambda_e - 1$	—	0
Cosmological Scale Factor	$a(t) = 1/(1+z)$	—	0
Hubble Parameter	$H = \dot{a}/a$	$\mathrm{s^{-1}}$	1

Remark

In the Heaviside-Lorentz system of natural units, we also set $\epsilon_0 = \mu_0 = 1$. Which one can hence get the dimension of charge q is 0.

1.2 4-Vectors and Tensors

1.2.1 4-Vectors

FOUNDATION

In special relativity, we treat the time and space coordinates on equal footing as a 4-vector $x^\mu = (t, x, y, z) = (x^0, x^1, x^2, x^3)$. or in other words, we used the 4-vector to combine the time and space coordinates together.

THEORY

By using the 4-vector, we can write the Lorentz transformation in-terms of the Lorentz transformation matrix and the 4-vector:
WLOG, we can consider a boost in the x-direction with velocity v,

$$\begin{pmatrix} t' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix},$$

where $\gamma = 1/\sqrt{1-v^2}$ is the Lorentz factor, and $\beta = v/c$ is the velocity in units of c ($\beta = v$ in natural units).

1.2.2 Einstein Summation Convention

PHYSICS INTUITION

Each time when we are doing the 4-vectors operations like the Lorentz transformation, we are doing the summation of the products of the components of the 4-vectors and the Lorentz transformation matrix.

It can be very tedious to write down the summation every time, so we want a more compact notation to represent the summation.

THEORY

Using the example of the Lorentz transformation above, if we want to do the transformation in traditional way, we have to write down the summation explicitly:

$$x^{\mu'} = \Sigma_{\nu=0}^{3} \Lambda^\mu{}_\nu x^\nu.$$

By using the Einstein summation convention, we can simply write down the transformation as:

$$x^{\mu'} = \Lambda^\mu{}_\nu x^\nu.$$

1.2.3 Metric Tensor

Physics Intuition

In a non-Euclidean space, such as the Minkowski spacetime in special relativity, the distance between two points is not simply what we seen in our eyes.
The spacetime interval between two events(or points) is no longer measured by the Pythagorean theorem, so we need to use another ruler to measure it.

In general, we used a metric tensor as the ruler to measure the distance of 2 point in a specific spacetime.

Theory

In traditional Euclidean space, the metric tensor is simply the identity matrix, so the distance between two points is given by the Pythagorean theorem:

$$ds^2 = \begin{pmatrix} dx & dy & dz \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} = dx^2 + dy^2 + dz^2.$$

In Minkowski spacetime, the metric tensor is given by:

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

so the spacetime interval is given by:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = -dt^2 + dx^2 + dy^2 + dz^2$$

or in other words, the spacetime interval is calculated by the dot product of the 4-vector with itself by using the metric tensor as the dot product operator:

Theory

$$A \cdot B = g_{\mu\nu} A^\mu B^\nu = \begin{pmatrix} A^0 & A^1 & A^2 & A^3 \end{pmatrix} \begin{pmatrix} g_{00} & g_{01} & g_{02} & g_{03} \\ g_{10} & g_{11} & g_{12} & g_{13} \\ g_{20} & g_{21} & g_{22} & g_{23} \\ g_{30} & g_{31} & g_{32} & g_{33} \end{pmatrix} \begin{pmatrix} B^0 \\ B^1 \\ B^2 \\ B^3 \end{pmatrix}$$

1.2.4 Tensors

Physics Intuition

Now you may wonder, why sometimes the index is up, and sometimes it is down? Is there any difference between them?
The answer is yes, they are different.

The upper index represents a **contravariant** component, while the lower index represents a **covariant** component.
By the combination of the **contravariant** and **convariant** components, we can construct a more general mathematical object called a **tensor**.

Foundation

In tensor calculus and differential geometry, the position of an index (upper or lower) indicates how the component of the tensor transforms when you change the coordinate system.

Direct Mapping:

- **Upper Index** (μ in V^μ): Represents a **Contravariant** component.
- **Lower Index** (μ in V_μ): Represents a **Covariant** component.

Transformation Rules:
The terms *contravariant* and *covariant* describe how components change relative to **basis vectors** when switching coordinates (e.g., Cartesian to Polar, or meters to centimeters).

Contravariant (Upper Index):

- **Symbol:** V^μ
- **Transformation:** Transforms **inversely** (against) the basis vectors.
- **Law:** $V'^\mu = \dfrac{\partial x'^\mu}{\partial x^\nu} V^\nu$
- **Intuition:** If your ruler shrinks (smaller basis vectors), the numerical value must grow to describe the same physical length. Example: 1 m = 100 cm.

Covariant (Lower Index):

- **Symbol:** V_μ
- **Transformation:** Transforms **in the same way** (with) the basis vectors.

- **Law:** $V'_\mu = \dfrac{\partial x^\nu}{\partial x'^\mu} V_\nu$

- **Intuition:** Like a gradient (slope). If the ruler shrinks, the slope per unit distance also shrinks.

Geometric Interpretation:

- **Contravariant Vectors** (V^μ): Standard "arrows" tangent to a curve, living in the **Tangent Space**.

- **Covariant Vectors** (V_μ): Called **Dual Vectors** or **One-Forms**. They are functions that map vectors to scalars and live in the **Cotangent Space**. Geometrically visualized as stacked parallel surfaces pierced by vectors.

Einstein Summation Convention:
The distinction is critical, repeated indices (one upper, one lower) indicate summation. Only these are coordinate-invariant scalars.

- **Valid:** $S = A^\mu B_\mu \quad \Rightarrow$ scalar invariant
- **Invalid:** $A^\mu B^\mu$ or $A_\mu B_\mu \quad \Rightarrow$ not coordinate-invariant (not summation)

Summary Table

Property	Contravariant	Covariant
Index Position	Upper ($^\mu$)	Lower ($_\mu$)
Notation	V^μ	V_μ
Also Called	Vector	Dual Vector / One-Form
Transformation	Opposite to basis	Same as basis
Example	Displacement, Velocity	Gradient
Metric Operation	Lowered by $g_{\mu\nu}$	Raised by $g^{\mu\nu}$

THEORY

Raising and Lowering Indices:
With the **Metric Tensor** ($g_{\mu\nu}$), you can convert between contravariant and covariant components:

- **Lowering:** $V_\mu = g_{\mu\nu} V^\nu$ (contravariant $\to$ covariant)
- **Raising:** $V^\mu = g^{\mu\nu} V_\nu$ (covariant $\to$ contravariant)

And the contravariant(inverse) metric $g^{\mu\nu}$ is the inverse of the covariant metric $g_{\mu\nu}$, so we have:

Theory

$$g^{\mu\nu} g_{\nu\sigma} = \delta^{\mu}{}_{\sigma},$$

where $\delta^{\mu}{}_{\sigma}$ is the Kronecker delta, which is 1 if $\mu = \sigma$ and 0 otherwise.

As a results, we can further simplified the operations like the dot product and trace:

Theory

$$A \cdot B = g_{\mu\nu} A^{\mu} B^{\nu} = A_{\nu} B^{\nu} = A^{\mu} B_{\mu}$$
$$T = g^{\mu\nu} T_{\mu\nu} = g_{\mu\nu} T^{\mu\nu} = T^{\nu}{}_{\nu} = T_{\mu}{}^{\mu}$$

Question Example 1.2: Trace of Spherical Minkowski Metric

Calculate the trace of the spherical Minkowski metric $g_{\mu\nu} = \mathrm{diag}(-1, 1, r^2, r^2 \sin^2\theta)$.

Solution. We need to first calculate the inverse metric $g^{\mu\nu}$, which is given by:

$$g^{\mu\nu} = g_{\mu\nu}^{-1} = \mathrm{diag}(-1, 1, r^{-2}, r^{-2} \sin^{-2}\theta).$$

Then we can calculate the trace:

$$T = g^{\mu\nu} g_{\mu\nu} = (-1)(-1) + (1)(1) + (r^{-2})(r^2) + (r^{-2} \sin^{-2}\theta)(r^2 \sin^2\theta) = 4.$$

Important Note

In flat Euclidean space with Cartesian coordinates, $g_{\mu\nu} = \delta_{\mu\nu}$ (identity), so $V^{\mu} = V_{\mu}$ numerically.

However, for other spacetimes (e.g., Minkowski, curved), the metric is not the identity, and V^{μ} and V_{μ} can differ significantly. The distinction becomes crucial in General Relativity and curvilinear coordinates. To be safe, treat them as different objects and always check the metric when raising/lowering indices.

EXTRA

The tensor T is said to be of type (p,q) if it has:

- p: The number of contravariant indices(upper indices)
- q: The number of convariant indices(lower indices)

Tensor with different types cannot be added together or equal to each other, in such a meaning, you can think of the type of a tensor as its "dimension" in the space of tensors, and only tensors with the same type can be compared or combined.

Tensor Rank: The total rank of a Tensor T is p + q (let said p + q = n), and we will said it is a rank-n-Tensor

Note that the rank of a tensor is not the same as linear algebra rank of a matrix. The rank of a tensor is the total number of indices, while the linear algebra rank is the maximum number of linearly independent rows or columns in a matrix. Thats mean a rank-2 tensor can have a linear algebra rank of 0, 1, or 2 depending on its components.

Tensor Type	p	q	Rank	Example
(0,0)	0	0	0	Scalar (e.g., Temperature)
(1,0)	1	0	1	Contravariant Vector (e.g., Displacement V^μ)
(0,1)	0	1	1	Covariant Vector (e.g., Gradient V_μ)
(1,1)	1	1	2	Mixed Tensor (e.g., Linear Transformation $\Lambda^\mu{}_\nu$)
(2,0)	2	0	2	Contravariant Tensor (e.g., Inverse Metric $g^{\mu\nu}$)
(0,2)	0	2	2	Covariant Tensor (e.g., Metric Tensor $g_{\mu\nu}$)
(1,3)	1	3	4	Riemann Curvature ($R^\rho_{\sigma\mu\nu}$)

1.2.5 Energy-Momentum Tensor

Physics Intuition

In Newtonian mechanics, we thought mass is the source of gravity, but in relativity, we found that mass is a form of energy($E = mc^2$), so energy must also be a source of gravity.
Further, momentum is also a form of energy($E^2 = p^2c^2 + m^2c^4$), so momentum must also be a source of gravity.

The energy-momentum tensor expand the concept of a scalar energy density to a 4x4 matrix that track of all "components" in a region of spacetime that contributes to gravity.

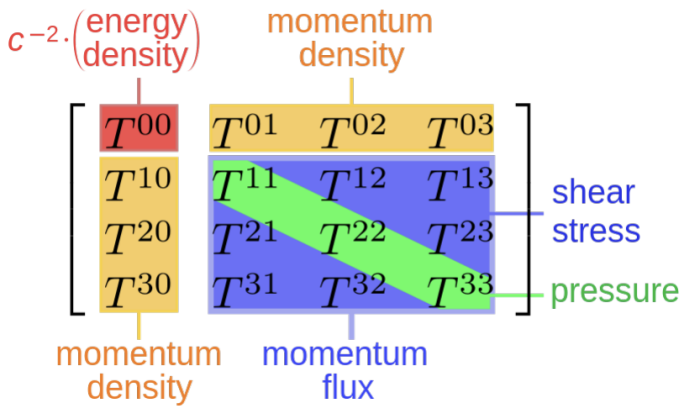

Figure 1.1: The components of the energy-momentum tensor $T^{\mu\nu}$ and their physical interpretations.

Foundation

Interpretation of Components:
For a component $T^{\mu\nu}$ of the energy-momentum tensor, it represents the flux of the μ-th component of the 4-momentum across a volume of constant ν-th coordinate.

(1) Time-Energy Component: T^{00} (Top-Left Corner)

- **What it is:** Energy density — the amount of energy per unit volume.
- **Includes:** Mass (via $E = mc^2$), heat, light, and chemical energy.
- **Gravitational role:** Primary source of gravity for everyday objects (e.g., the Earth).

(2) Energy Flux Components: T^{01}, T^{02}, T^{03} (Top Row)

- **What they measure:** Energy flowing through surfaces in the x-, y-, and z-directions.
- **Physical picture:** A beam of light passing through the volume carries energy in a direction.
- **Key insight:** In relativity, energy flux is equivalent to momentum density. Moving energy carries momentum!

(3) Momentum Density Components: T^{10}, T^{20}, T^{30} (Left Column)

- **What they measure:** The amount of momentum stored inside the volume.
- **Surprising fact:** These values are typically equal to the energy flux components above!
- **Implication:** Energy flow and momentum density are intimately related.

(4) Stress and Pressure Components: T^{11} to T^{33} (Bottom-Right 3×3 Grid)

- **Physical meaning:** Forces and stresses acting within the material.
- **Diagonal terms** (T^{11}, T^{22}, T^{33}): **Pressure**
  - Gas molecules bouncing off the walls of the volume exert pressure.
  - **Remarkable fact:** Pressure creates gravity!
  - A box of hot gas (high pressure) has slightly more gravitational mass than cold gas with the same rest mass, because pressure contributes to the stress-energy tensor.
- **Off-diagonal terms** (T^{12}, T^{23}, etc.): **Shear Stress**
  - Internal friction between different parts of the material.
  - Analogy: Sliding the top half of a deck of cards sideways relative to the bottom.

FOUNDATION

The energy-momentum tensor has 2 important properties:

(1) Symmetry: $T^{\mu\nu} = T^{\nu\mu}$

- Reason 1: Energy flux and momentum density are equal, so $T^{0i} = T^{i0}$.
- Reason 2: The stress tensor (bottom-right 3×3) is symmetric due to the conservation of angular momentum. If there were a torque, it would cause the material to rotate infinitely fast, which would violate the conservation of angular momentum.

(2) Conservation: $\partial_\mu T^{\mu\nu} = 0$

imagine a small box in space-time. The conservation law states that the amount of energy-momentum flowing into the box along the μ-th direction must equal the amount flowing out along the ν-th direction. This ensures that energy and momentum are conserved locally.

IMPORTANT NOTE

The energy-momentum tensor is a symmetric so there are only 10 independent components at a single point in spacetime.

If we further consider the conservation law, we can reduce the number of degree of freedom to 6, as it provides 4 additional constraints on the components of the energy-momentum tensor.

(i.e. 10 independent components, with 4 constraints from conservation, leaves 6 degrees of freedom.)

REMARK

In fact, for different type of fluids, the can have different number of degree of freedom, 6 is just the upper bound for a general fluid. For example, a perfect fluid has only 5 degree of freedom, which are the energy density(1) and pressure(1) and velocity(3), as it is isotropic and has no shear stress.

Theory

For frictionless perfect fluid, the energy-momentum tensor can be written as:

$$T^{\mu\nu} = (\rho + P)u^\mu u^\nu + Pg^{\mu\nu},$$

where ρ is the energy density, P is the pressure, and u^μ is the 4-velocity of the fluid.
WLOG, we can consider a comoving frame where the fluid is at rest, so $u^\mu = (1, 0, 0, 0)$, then we can simplify the energy-momentum tensor to:

$$T^{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{pmatrix}$$

The perfect fluid is a good approximation for many astrophysical and cosmological systems, such as stars, galaxies, and the large-scale structure of the universe. As it is isotropic and has no shear stress, it can be used to model the behavior of matter in these systems without having to consider the complexities of viscosity and heat conduction.

1.3 Einstein Field Equations

1.4 Cosmological Principle

The cosmological principle is a fundamental assumption in cosmology that states that the universe is homogeneous and isotropic on large scales. This means that the universe looks the same in all directions (isotropic) and has the same properties at every point (homogeneous) when viewed on sufficiently large scales.

Extra

The cosmic Microwave Background (CMB) radiation provides strong evidence for the cosmological principle, as it is observed to be nearly uniform in all directions, with only tiny fluctuations at the level of one part in 100,000.

CHAPTER 2

Hubble's Law and Cosmological Redshift (lecture 3-4)

With the fundamental knowledge in last chapter, we can now start to build up our universe model.

2.1 Robertson-Walker Metric

2.1.1 Metric for a Homogeneous and Isotropic Universe

PHYSICS INTUITION

In general relativity, the geometry of spacetime can be curved.
Therefore, we need another metric instead of the Minkowski metric to describe the geometry of the universe.
The Robertson-Walker metric was hence introduced to describe a homogeneous and isotropic universe in large scales.

THEORY

By the cosmological principle, the universe is homogeneous and isotropic on large scales.
In mathematics, there are only three possible spatial geometries that satisfy these conditions, defined by the constant curvature k:

- $k = 0$: flat geometry (Euclidean space)
- $k > 0$: closed geometry (spherical space)
- $k < 0$: open geometry (hyperbolic space)

And we need to write the line element dl^2 for such a spacetime.
Note that the cosmological principle only states that the universe is homogeneous and isotropic, but it does not forbid the universe to expand or contract.
Therefore, we need to introduce a time-dependent scale factor $a(t)$ to describe the expansion of the universe.

$$ds^2 = -dt^2 + a^2(t)dl^2$$

To fullfill the isotropy condition, we will use spherical coordinates to write the spatial line element dl^2:

$$dl^2 = g_{rr}dr^2 + g_{\theta\theta}d\theta^2 + g_{\phi\phi}d\phi^2$$

FOUNDATION

you may ask why we dont need to consider the cross terms like $g_{r\theta}drd\theta$ in the line element.

The reason is that the cross terms would break the isotropy of the universe. For example, if we have a cross term like $g_{r\theta}drd\theta$, it would mean that the radial distance between two points in the universe would depend on the direction we are looking at, which would violate the isotropy condition.

Therefore, we can safely ignore the cross terms in the line element when we are considering a homogeneous and isotropic universe.

THEORY

Flat geometry ($k = 0$):
It is trivial to get the line element for flat geometry, which is just the Euclidean space:

$$dl^2 = dr^2 + r^2 d\Omega^2 = dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)$$

which $d\Omega^2$ is the line element of a unit sphere(r = 1).
Hence, the metric for a flat universe is:

$$ds^2 = -dt^2 + a^2(t)[dr^2 + r^2 d\Omega^2] = -dt^2 + a^2(t)\left[dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)\right]$$

However, for the closed and open geometry, the coefficient of $d\Omega^2$ is not r^2, but a function of r.

Closed geometry ($k > 0$):
To get the line element for closed geometry like spherical space, we first consider a 3-sphere embedded in a 4-dimensional Euclidean space, which can be described by the equation:

$$x^2 + y^2 + z^2 + w^2 = R^2$$

where R is the radius of the 3-sphere.
We can then use spherical coordinates to parameterize the 3-sphere:

$$\begin{cases} x = R\sin\chi\sin\theta\cos\phi \\ y = R\sin\chi\sin\theta\sin\phi \\ z = R\sin\chi\cos\theta \\ w = R\cos\chi \end{cases}$$

where χ is the radial coordinate that ranges from 0 to π.
And we can get the metric components:

$$\begin{cases} g_{\chi\chi} = \frac{\partial}{\partial\chi}(x,y,z,w) \cdot \frac{\partial}{\partial\chi}(x,y,z,w) = R^2 \\ g_{\theta\theta} = \frac{\partial}{\partial\theta}(x,y,z,w) \cdot \frac{\partial}{\partial\theta}(x,y,z,w) = R^2 \sin^2\chi \\ g_{\phi\phi} = \frac{\partial}{\partial\phi}(x,y,z,w) \cdot \frac{\partial}{\partial\phi}(x,y,z,w) = R^2 \sin^2\chi \sin^2\theta \end{cases}$$

The line element for the 3-sphere can then be derived as:

$$dl^2 = R^2 \left[d\chi^2 + \sin^2\chi (d\theta^2 + \sin^2\theta d\phi^2) \right]$$

Hence, the metric for a closed universe is:

$$ds^2 = -dt^2 + a^2(t) R^2 \left[d\chi^2 + \sin^2\chi (d\theta^2 + \sin^2\theta d\phi^2) \right]$$

Open geometry ($k < 0$):
To get the line element for open geometry like hyperbolic space, we can consider a 3-hyperboloid embedded in a 4-dimensional Minkowski space, which can be described by the equation:

$$-x^2 - y^2 - z^2 + w^2 = R^2$$

where R is the radius of curvature of the hyperbolic space.
We can then use hyperbolic coordinates to parameterize the 3-hyperboloid:

$$\begin{cases} x = R \sinh\chi \sin\theta \cos\phi \\ y = R \sinh\chi \sin\theta \sin\phi \\ z = R \sinh\chi \cos\theta \\ w = R \cosh\chi \end{cases}$$

where χ is the radial coordinate that ranges from 0 to ∞.
And we can get the metric components:

$$\begin{cases} g_{\chi\chi} = \frac{\partial}{\partial\chi}(x,y,z,w) \cdot \frac{\partial}{\partial\chi}(x,y,z,w) = R^2 \\ g_{\theta\theta} = \frac{\partial}{\partial\theta}(x,y,z,w) \cdot \frac{\partial}{\partial\theta}(x,y,z,w) = R^2 \sinh^2\chi \\ g_{\phi\phi} = \frac{\partial}{\partial\phi}(x,y,z,w) \cdot \frac{\partial}{\partial\phi}(x,y,z,w) = R^2 \sinh^2\chi \sin^2\theta \end{cases}$$

The line element for the 3-hyperboloid can then be derived as:

$$dl^2 = R^2 \left[d\chi^2 + \sinh^2\chi (d\theta^2 + \sin^2\theta d\phi^2) \right]$$

Hence, the metric for an open universe is:

$$ds^2 = -dt^2 + a^2(t) R^2 \left[d\chi^2 + \sinh^2\chi (d\theta^2 + \sin^2\theta d\phi^2) \right]$$

2.1.2 Robertson-Walker metric

Physics Intuition

In order to be simplified, we want to make a unified form of the metric for all three types of geometry.

Theory

Unified form:

$$\begin{pmatrix} g_{tt} & 0 & 0 & 0 \\ 0 & g_{rr} & 0 & 0 \\ 0 & 0 & g_{\theta\theta} & 0 \\ 0 & 0 & 0 & g_{\phi\phi} \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{a^2(t)}{1-kr^2} & 0 & 0 \\ 0 & 0 & a^2(t)r^2 & 0 \\ 0 & 0 & 0 & a^2(t)r^2\sin^2\theta \end{pmatrix}$$

where k is the curvature constant that can take values of 0, 1, or -1 for flat, closed, and open geometries, respectively.
And we defined the r:

$$r = \begin{cases} \chi & ,k = 0 \\ \sin\chi & ,k > 0 \\ \sinh\chi & ,k < 0 \end{cases}$$

Foundation

By the substitution of r into the unified metric and letting the unit $R = 1$, you can easily to get the $d\theta^2$ and $d\phi^2$ terms of 3 types of geometries, and get the corresponding dr^2 term by using the chain rule:

$$dr^2 = \begin{cases} d\chi^2 & ,k = 0 \\ \cos^2\chi d\chi^2 & ,k > 0 \\ \cosh^2\chi d\chi^2 & ,k < 0 \end{cases}$$

$$\Rightarrow \frac{dr^2}{1-kr^2} = \begin{cases} d\chi^2 & ,k = 0 \\ d\chi^2 & ,k > 0 \\ d\chi^2 & ,k < 0 \end{cases}$$

And this unified metric is called the **Robertson-Walker metric**, which can be written as:

$$ds^2 = -dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2(d\theta^2 + \sin^2\theta d\phi^2)\right]$$

Remark

In modern cosmology, we will use Friedmann-Lemaître-Robertson-Walker (FLRW) metric, but they are essentially the same, the FLRW metric derived from the Einstein field equations instead of geometry approach like the Robertson-Walker metric, and is used for explaining the dynamics of the universe, while the Robertson-Walker metric is used for describing the geometry of the universe.

2.1.3 the Scale Factor $a(t)$

Physics Intuition

Now we get the Robertson-Walker metric, and we know that through "time" the scale factor $a(t)$ can change, and the metric can be different at different time, which means the proper distance between two points in the universe can change with time.

But what the "time" here means? Isn't the special relativity tells us that there is no absolute time?
The "time" here is the proper time of comoving observers, which are observers that are at rest with respect to the expansion of the universe.
For comoving observers, the proper time τ is the same as the coordinate time t.

Physics Intuition

As the scale factor $a(t)$ applied on the metric instead of the coordinates difference dr, $d\theta$, and $d\phi$, we can think the measurement of coordinate spatial distance is fixed, but the measurement unit (the ruler itself) is changing with time.

$$\Delta D_{proper} = a(t)\Delta D_{coordinate}$$

For example, if we have 2 points on balloon, we draw the marks on the balloon to measure the distance between the two points, when the balloon is expanding, the number of marks between the two points is fixed, but the distance between each mark is increasing with time, which means the proper distance between the two points is increasing with time.

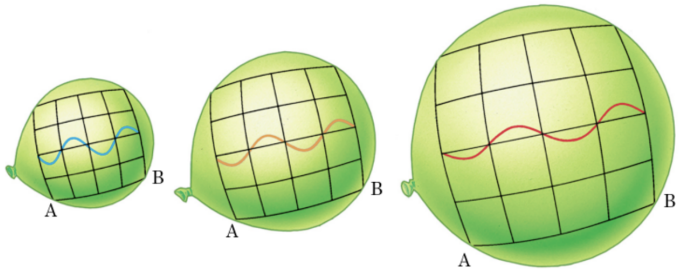

Figure 2.1: The balloon analogy for the expansion of the universe.

Remark

Although scale factor $a(t)$ can change with time, we assume that the curvature constant k does not change with time, which means the geometry of the universe is fixed.

2.2 Hubble's Law

2.2.1 Redshift and Recession Velocity

CHAPTER 3

Cosmic Evolution (Lecture 5-7)

CHAPTER 4

The Standard Model (Lecture 8)

Particle physics is not the main focus of this course, but it is important to have a basic understanding of the fundamental particles and interactions that govern the universe, as they play a crucial role in cosmology, especially in the early universe.

4.1 The Standard Model of Particle Physics

PHYSICS INTUITION

The Standard Model is a theoretical framework that describes the fundamental particles and their interactions, except for gravity. The fundamental particles are the smallest building blocks we have know so far, that build up matter(and energy) of the universe, and the interactions between them govern the behaviour of all physical phenomena.

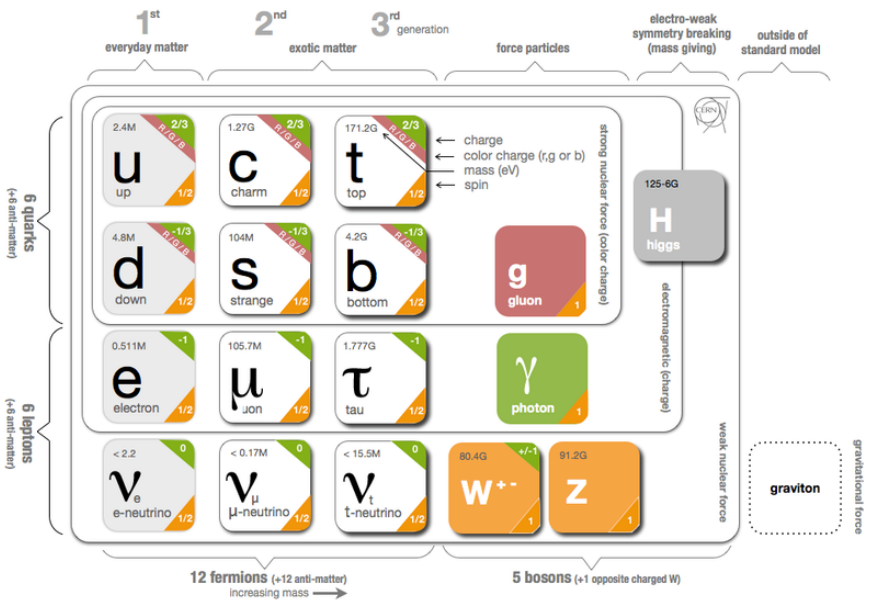

Figure 4.1: The Standard Model of particle physics, showing the fundamental particles and their interactions.

FOUNDATION

From the above figure, we can see that for each particles, they have below properties:

- **Spin**: This is an intrinsic form of angular momentum that particles possess. It determines the statistics that particles obey (Fermi-Dirac for fermions and Bose-Einstein for bosons) and their behaviour under rotations.
  - **Spin 0 (scalar)**: e.g. Higgs boson.
  - **Spin 1/2 (fermions)**: e.g. electrons, quarks.
  - **Spin 1 (vector)**: e.g. photon, W and Z bosons.
- **Charge**: This is a fundamental property that determines how particles interact with electromagnetic fields. Particles can have positive, negative, or zero charge. The same type of particles with different charges are said as particles-antiparticles pair, such as electron and positron, proton and antiproton, etc.
- **Mass**: This is a measure of the amount of matter in a particle. It determines how particles respond to gravitational fields and how they interact with the Higgs field.
- **Color Charge**: This is a property of quarks that determines how they interact with the strong force. Each quark can have 1 of 3 color charges (red, green, blue), and the strong force is mediated by gluons that carry 1 of the 8 color charges.
- **Chiralility**: This describes the handedness of particles, which is important for weak interactions. Left-handed particles and right-handed particles can have different interactions with the weak force.

  For a group of particles with similar properties, we can classify them into a family, such as the three **generations** of fermions (electron, muon, tau and their corresponding neutrinos; up, down, charm, strange, top, bottom quarks), and the gauge bosons (photon, W and Z bosons, gluons).
  Sometimes, we will also use the term **flavour** to refer to the different types of particles within a family, such as the different flavours of quarks (up, down, charm, strange, top, bottom) and leptons (electron, muon, tau).
- **Interactions**: The fundamental particles interact with each other through four fundamental forces: gravity, electromagnetism, weak nuclear force, and strong nuclear force. The Standard Model describes the electromagnetic, weak, and strong interactions, but it does not include gravity.

FOUNDATION

The Standard Model consists of three main components:

- **Fermions**: These are the matter particles, which include quarks and leptons. Quarks combine to form protons and neutrons, while leptons include electrons and neutrinos.
- **Gauge Bosons**: These are the force carriers that mediate the fundamental interactions. They include photons (electromagnetic force), W and Z bosons (weak force), and gluons (strong force).
- **Higgs Boson**: This particle is responsible for giving mass to other particles through the Higgs mechanism.

FOUNDATION

Due to the strong interaction is asymptotically free, the quarks usually exist in forms of hadrons with 2 types:

- **Baryons**: These are made up of three quarks, and are subatomic particles such as protons $(p = uud)$ and neutrons $(n = udd)$.
- **Mesons**: These are made up of a quark and an antiquark, such as pions (π^+, π^0, π^-) and kaons (K^+, K^0, K^-).

Note not all quarks are confined in hadrons, such as the top quark, which is too heavy to form hadrons before it decays.

REMARK

The Standard Model(SM) just a zoo of fundamental particles discovered so far, but it is not a complete theory of everything. It does not include gravity, and it cannot explain dark matter, dark energy, or the matter-antimatter asymmetry in the universe. Therefore, physicists are still searching for a more fundamental theory that can unify all forces and particles in the universe, which is often referred to as the "Theory of Everything(TOE)".

4.2 Higgs Mechanism and Mass Generation

Physics Intuition

For the boson particles, such as W and Z bosons, they are massive, but if we want to write down a gauge invariant Lagrangian for them, we cannot include a mass term for them, as it will break the gauge symmetry.

In a nutshell, allow these force carriers to have mass would means they are moving slower than the speed of light, which provide a extra degree of freedom for them; The extra degree of freedom provides extra force(e.g. for W and Z bosons, it is the weak force), which causing infinite force at high energy, causing the theory broken.

Important Note

One of the common mistakes is that people think the Higgs mechanism gives mass to all particles, but in fact, it only play a dominant role in giving mass to the W and Z bosons;

while for fermions, such as quarks, only $\sim 1\%$ of their mass come from the Higgs mechanism, the rest $\sim 99\%$ of their mass come from the strong interaction that bind them together in hadrons, such as protons and neutrons.

Physics Intuition

The Higgs mechanism is a process by which particles acquire mass through their interactions with the Higgs field. The Higgs field is a scalar field that permeates all of space, and it has a non-zero vacuum expectation value (VEV). When particles interact with the Higgs field, they acquire mass proportional to the strength of their interaction with the field.

On a easier-to-understand level, we can think of the Higgs field as a kind of "cosmic molasses" that fills the universe. When particles move through this molasses, they experience resistance, which manifests as mass. The stronger the interaction with the Higgs field, the more resistance they experience, and thus the greater their mass.

For example, the W and Z bosons acquire mass through their interactions with the Higgs field, while photons remain massless because they do not interact with the Higgs field.

In Higgs mechanism, the mass property of gauge bosons no longer come from them selves(like the fermions), but come from the interaction with the Higgs field.

By this tricks, the gauge bosons extra degree of freedom is provided by the Higgs field, thus allow the gauge bosons seems to be "massive" without breaking the gauge symmetry.

THEORY

the total number of fundamental degrees of freedom in phase transition with the Higgs mechanism:

$$N_{\text{dof}} = 2 \times N_G + N_{\text{Higgs}} = 3 \times N_G + (N_{\text{Higgs}} - N_G)$$

where N_G is the number of gauge bosons that acquire mass through the Higgs mechanism, and N_{Higgs} is the number of Higgs fields. The first term $2 \times N_G$ accounts for the two transverse polarizations of the gauge bosons, while the second term $(N_{\text{Higgs}} - N_G)$ accounts for the remaining degrees of freedom from the Higgs fields after giving mass to the gauge bosons.

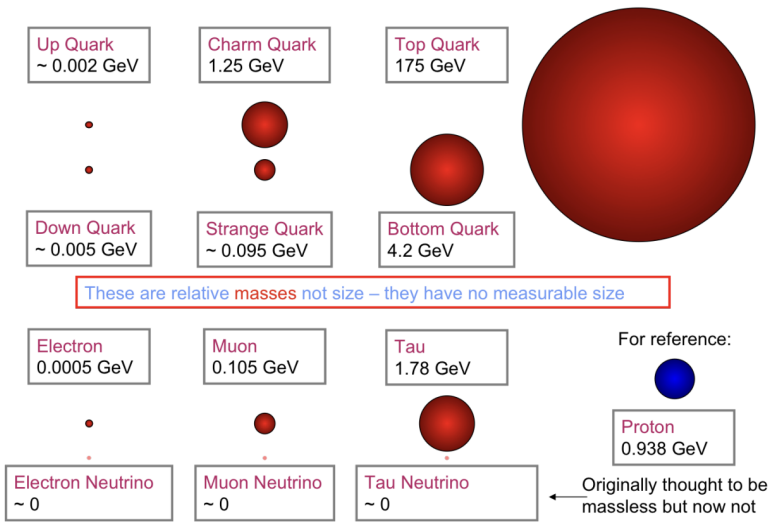

Figure 4.2: Some examples of masses of fundamental particles in the Standard Model.

CHAPTER 5

Thermodynamics and Primordial Soup (Lecture 9-12)

To get the factors of energy density of radiation and matter, so that we can know the Evolution of the universe in a more general way, we will need to find out how radiation and matter mixture evolves.

Thermodynamics allow us to know the contribution of different particle species to the total energy density of the universe, and thus determine the expansion rate and thermal history of the universe.

5.1 Statistical Distributions

5.1.1 Energy-Momentum Relation

To calculate the value of various thermodynamic quantities, we need first clarify some basic concepts and notations first for convenience.

In below contents, we will use the $E(p)$ notation to represent the energy of a particle with momentum p and mass m.

FOUNDATION

Recall special relativity, the energy of a particle with mass m and three-momentum p is given by

$$E(\vec{p}) = \sqrt{|\vec{p}|^2c^2 + m^2c^4}.$$

Taken $c = 1$ and $p = |\vec{p}|$, we have $E(p) = \sqrt{p^2 + m^2}$.

In a Non-relativistic limit ($p \ll m$):

$$E(p) = m\sqrt{1 + \frac{p^2}{m^2}} \approx m + \frac{p^2}{2m}.$$

In an Ultra-relativistic limit ($p \gg m$):

$$E(p) \approx p.$$

5.1.2 Chemical Potential

FOUNDATION

μ is the chemical potential, which measures the change in internal energy when a particle is added to the system at constant entropy and volume.

For ideal gases, $\mu \to 0$, and we will assume $\mu = 0$ in below contents unless otherwise specified.

REMARK

The chemical potential μ is typically zero for photons and other massless bosons, while it can be non-zero for massive particles depending on the context (e.g., in a gas of electrons).

while for particles and their own anti-particles, $\mu_+ = -\mu_-$.

5.1.3 Thermal Equilibrium Distributions

FOUNDATION

For a state with energy E, and particle number N,

THEORY

In thermal equilibrium, the distribution of particle energies follows specific statistical distributions:

- **Fermi-Dirac distribution** for fermions:

$$f(p) = \frac{1}{e^{(E(p)-\mu)/k_B T} + 1}$$

- **Bose-Einstein distribution** for bosons:

$$f(p) = \frac{1}{e^{(E(p)-\mu)/k_B T} - 1}$$

- **Maxwell-Boltzmann distribution** for classical particles:

$$f(p) = e^{-(E(p)-\mu)/k_B T}$$

In general, the distribution can be written as:

$$f(p) = \frac{1}{e^{(E(p)-\mu)/k_B T} \pm 1},$$

where the $+$ sign is for fermions and the $-$ sign is for bosons.

THEORY

In non-relativistic limit(low temperature & low density), the Fermi-Dirac and Bose-Einstein distributions reduce to the Maxwell-Boltzmann distribution:

$$e^{(E(p)-\mu)/k_B T} \gg 1 \implies f(p) \approx e^{-(E(p)-\mu)/k_B T}$$

5.2 Number Density, Energy Density, and Pressure

5.2.1 Spin Degeneracy Factor

In below contents, we define g as Spin degeneracy factor(spin states), which counts the number of internal degrees of freedom (e.g., spin states) for a particle species.

FOUNDATION

g

EXTRA

The g values for some common particles are:

Particle	symbol	g
Photon	g_γ	2
Electron/Positron	g_e^-, g_e^+	2
Z boson	g_Z	3
Higgs boson	g_H	1

5.2.2 Derivation of the 3 equations

FOUNDATION

THEORY

The number density n, energy density ρ, and pressure P of some particle i are given by:

$$n_i = \frac{g_i}{(2\pi)^3} \int f_i(p)\, d^3p,$$

$$\rho_i = \frac{g_i}{(2\pi)^3} \int E(p) f_i(p)\, d^3p,$$

$$P_i = \frac{g_i}{(2\pi)^3} \int \frac{p^2}{3E(p)} f_i(p)\, d^3p$$

FOUNDATION

By substituting the Fermi-Dirac, Bose-Einstein, or Maxwell-Boltzmann distribution functions into the above integrals, we can derive explicit expressions for n_i, ρ_i, and P_i for different particle species under various conditions (e.g., relativistic vs. non-relativistic limits).

(1) Relativistic Fermions ($E \approx p$):

- Number density: $n_i = \frac{3\zeta(3)}{4\pi^2} g_i T^3$
- Energy density: $\rho_i = \frac{7\pi^2}{120} g_i T^4$
- Pressure: $P_i = \frac{1}{3}\rho_i$

(2) Relativistic Bosons ($E \approx p$):

- Number density: $n_i = \frac{\zeta(3)}{\pi^2} g_i T^3$
- Energy density: $\rho_i = \frac{\pi^2}{30} g_i T^4$
- Pressure: $P_i = \frac{1}{3}\rho_i$

(3) Non-relativistic Particles ($E \approx m + p^2/2m$):

- Number density: $n_i = g_i \left(\frac{m_i T}{2\pi}\right)^{3/2} e^{-m_i/T}$
- Energy density: $\rho_i = n_i m_i$
- Pressure: $P_i = n_i T$

THEORY

Number density, energy density, and pressure for particles with different thermal equilibrium distributions:

	Relativistic Fermions	Relativistic Bosons	Non-relativistic Particles (Fermions/ Bosons)
Number Density n_i	$\frac{3}{4}\frac{\zeta(3)}{\pi^2} g_i T^3$	$\frac{\zeta(3)}{\pi^2} g_i T^3$	$g_i \left(\frac{m_i T}{2\pi}\right)^{3/2} e^{-m_i/T}$
Energy Density ρ_i	$\frac{7}{8}\frac{\pi^2}{30} g_i T^4$	$\frac{\pi^2}{30} g_i T^4$	$n_i m_i$
Pressure P_i	$\frac{1}{3}\rho_i$	$\frac{1}{3}\rho_i$	$n_i T$

where $\zeta(3) \approx 1.202$ is the Riemann zeta function evaluated at 3.

We can see once the particles no longer relativistic, the number density and energy density will be exponentially suppressed.
It is because in the thermal plasma, the energy of particle are $E \sim T$, and when $T \ll 2m$, the probability of pair production which require energy $E \geq 2m$ will be very low, thus the number density and energy density will be suppressed.

Physics Intuition

It also explains why very few positrons exist in the current universe!
In the early universe, $T \gg m_e \approx 0.511\,\text{MeV}$, e^+e^- pairs were abundant.
As the universe cooled below $T \sim m_e$, the number density of positrons (and electrons) dropped exponentially, leaving only a tiny fraction that survived annihilation, which is why we observe so few positrons today.

Theory

We can see the relationship between p and ρ for different relativistic limits:

$$P = \begin{cases} \frac{1}{3}\rho & , T \gg m \\ nT \ll \rho & , T \ll m \end{cases}$$

In fact, it also consistent with the assumption of state parameter of radiation and matter, which are $w = 1/3$ and $w = 0$ when in relativistic and non-relativistic limits respectively.

Theory

The average particle energy is given by:

$$\langle E \rangle = \frac{\rho_i}{n_i} = \begin{cases} \frac{7\pi^4}{180\zeta(3)} T \approx 3.151T & , T \gg m \text{ for relativistic fermions} \\ \frac{\pi^4}{30\zeta(3)} T \approx 2.701T & , T \gg m \text{ for relativistic bosons} \\ m + \frac{3}{2}T & , T \ll m \text{ for non-relativistic particles} \end{cases}$$

Thats mean we can have a simplification of $\langle E \rangle \sim T$ for relativistic particles, as they have same order of magnitude, which is low enough error in astronomy :)

5.3 Effective Number of Relativistic DOF

Theory

The Friedmann equation shows that the expansion rate of the universe is determined by the total energy density ρ:

$$\rho(T) = \sum_i \rho_i(T)$$

where i runs over all particle species in the thermal plasma.

Considering in the the early universe, energy density is dominated by relativistic particles, we can directly sum over the relativistic species, and define the effective number of relativistic degrees of freedom g_* as:

$$\rho(T) = \frac{\pi^2}{30} g_*(T) T^4,$$

where:

$$g_*(T) = \sum_{\text{bosons}} g_i + \frac{7}{8} \sum_{\text{fermions}} g_i$$

Important Note

Note the above equation is only valid in early universe which radiation dominates. When universe cools down and matter dominates, the energy density will be dominated by non-relativistic particles, and above equation will no longer valid. In that case, we need to consider the contribution of non-relativistic particles to the energy density, which is given by $\rho_i = n_i m_i$.

Remark

g_* is depends on T(i.e. $g_*(T)$), as the sum runs over only relativistic species, with $m_i \ll T$. As the universe cools down, more and more species become non-relativistic and drop out of the sum, thus g_* decreases with time.

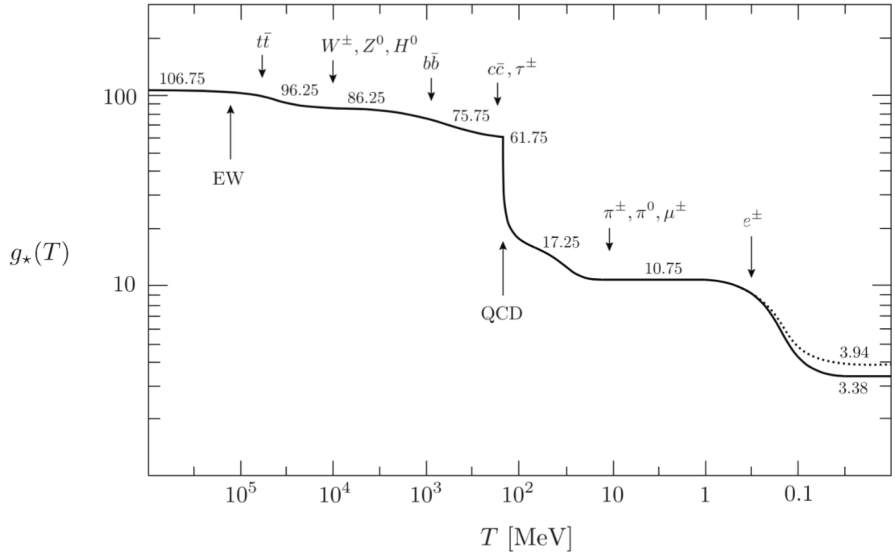

Figure 5.1: Evolution of g_* with temperature.

Extra

In fact, the above equation of g_* assume that all particles are not decoupled and in same temperature, to be more general, we need to consider the contribution of decoupled particles with different temperature, and the equation will be:

$$g_*(T) = \sum_{\text{bosons}} g_i \left(\frac{T_i}{T}\right)^4 + \frac{7}{8} \sum_{\text{fermions}} g_i \left(\frac{T_i}{T}\right)^4$$

5.4 Entropy

5.4.1 Definition of Entropy

Physics Intuition

Far from above topic, we only discuss the macrostate of a system(e.g., temperature, pressure, volume), but we haven't discussed the microstate of a system, which is the specific configuration of particles(i.e. the positions and momenta of every particles) that make up the system.

In a given macrostate, there can be many different microstates that correspond to the same macrostate. For example, a macrostate of 2 coins combined with 1 H and 1 T can be achieved by 2 different microstates: HT and TH. While a

macrostate of 2 coins combined with 2 H can only be achieved by 1 microstate: HH.

The entropy of a system is a measure of the number of microstates that correspond to a given macrostate. The more microstates there are, the higher the entropy.

In the context of thermodynamics, the concentrated energy or particles(mass) in a small region of space has low entropy, while the dispersed energy or particles in a large region of space has high entropy.

FOUNDATION

State Function is function of state variables, which only depends on the current state of the system, and is independent of the path taken to reach that state. Below are some examples of state functions:

- Internal Energy (U)
- Volume (V)
- Particle Number (N)
- Temperature (T)

And the Entropy (S) is also a state function.

REMARK

The entropy value is not absolute, but relative. We can only measure the change of entropy ΔS between two states, but not the absolute value of entropy S itself.

This is because the absolute value of entropy depends on the choice of reference state, which is arbitrary. For example, we can choose the reference state to be the state of zero entropy, which corresponds to a perfectly ordered system with only one microstate.

In that case, the entropy of any other state will be positive, and the change in entropy ΔS will be equal to the absolute value of entropy S for that state. However, if we choose a different reference state, the absolute value of entropy will be different, but the change in entropy ΔS between two states will remain the same, as it only depends on the difference in the number of microstates between the two states, which is independent of the choice of reference state.

FOUNDATION

There is 2 different definitions of entropy depending on the context:

- **classical thermodynamics definition:**

$$dS = \frac{dQ}{T}$$

where dQ is the heat added to the system, and T is the temperature of the system.

It is based on the idea below:
For a system, if we add heat(dQ), the system gain more energy, and particles in the system will have more energy to move around, which allow them to access more velocities microstates. Thus, the number of microstates accessible to the system will increase, which means the entropy of the system will increase.

$$dS \propto dQ$$

On the other hand, if the system has higher temperature, the average energy of particles in the system already high, and their already larger number of microstates accessible to them. Thus, adding the same amount of heat dQ to a system with higher temperature will result in a smaller increase in the number of microstates, and thus a smaller increase in entropy.

$$dS \propto \frac{1}{T}$$

- **statistical mechanics definition:**

$$S = k_B \ln \Omega$$

where k_B is the Boltzmann constant, and Ω is the number of microstates corresponding to the macrostate of the system.

It is based on the idea below:
Entropy(S) has extensive property, which means that if we have two independent systems A and B, the total entropy of the combined system is the sum of the entropies of the individual systems:

$$S_{\text{total}} = S_A + S_B$$

On the other hand, the number of microstates(Ω) has multiplicative property, which means that if we have two independent systems A and B, the total

number of microstates of the combined system is the product of the number of microstates of the individual systems:

$$\Omega_{\text{total}} = \Omega_A \cdot \Omega_B$$

Therefore, to fulfill both properties, we can define the entropy as the logarithm of the number of microstates:

$$S \propto \ln \Omega$$

Now we can use the above definition of entropy to prove the 2nd law of thermodynamics.

FOUNDATION

Start from Carnot cycle, assume we have a reversible carnot engine operating between two heat reservoirs at temperatures T_H and T_C ($T_H > T_C$).

$$\text{Efficiency}\eta = 1 - \frac{T_C}{T_H} = 1 - \frac{Q_C}{Q_H}\frac{Q_C}{T_C} = \frac{Q_H}{T_H}$$

In a complete cycle:

$$\oint_{\text{rev}} \frac{dQ}{T} = 0$$

Note carnot cycle is theoretical most efficient engine, which means for any cycle:

$$\oint \frac{dQ}{T} \leq 0$$

And for a heat loss irreversible process, we have:

$$\oint_{\text{irr}} \frac{dQ}{T} < 0$$

So only if the process is reversible, the change in entropy will be equal to the integral of dQ/T;

And if the process is irreversible, the change in entropy will be greater than the integral of dQ/T, or equivalently, the integral of dQ/T will be less than the change in entropy due to the heat loss in the irreversible process.

THEORY

The 2nd law of thermodynamics can been expressed by the general entropy inequality:

$$\Delta S = S_{\text{final}} - S_{\text{initial}} \geq \int_{\text{initial}}^{\text{final}} \frac{dQ}{T}|_{\text{path}}$$

where ΔS is the change in entropy of the system,
If the process is reversible, then lhs is $=$ to rhs;
and if the process is irreversible, then lhs is $>$ than rhs.

In an isolated system, there is no heat exchange with the surroundings, so $dQ = 0$, thus the inequality reduces to:

$$\Delta S_{\text{isolated}} \geq 0$$

and we get:

$$\begin{cases} \Delta S = 0 & \text{, for reversible processes} \\ \Delta S > 0 & \text{, for irreversible processes} \end{cases}$$

PHYSICS INTUITION

According to the second law of thermodynamics, the total entropy of an isolated system can never decrease over time, which means that the universe tends to evolve towards states with higher entropy.

In the macro level, it shows the universe tends to spread out energy and matter, and become more homogeneous and isotropic over time.

5.4.2 Conservation of Entropy

PHYSICS INTUITION

Assuming the universe is an isolated system (adiabatic), by 2nd law of thermodynamics, the total entropy of the universe is either constant or increasing over time.

Further, if we assume the evolution of the universe is quasi-static and reversible, thats mean the universe evolves through a series of equilibrium states, and there is no heat exchange with the surroundings, so the ΔS of the universe is zero, and entropy is conserved.

Theory

The fundamental thermodynamic relation is given by:

$$dU = TdS - PdV + \mu dN$$

by rearranging the above equation, we can get:

$$dS = \frac{dU + PdV - \mu dN}{T}$$

or

$$s = \frac{\rho + p}{T},$$

where $s = \frac{dS}{dV}$, $\rho = \frac{dU}{dV}$, and p are the equilibrium entropy density, energy density, and pressure of the system respectively.

In a more general case, considering $U = \rho V$ and $V \propto a^3$, and assuming $\mu = 0$, we can get:

$$\begin{aligned} dU &= Tds - pdV \\ d\rho V + \rho dV &= Tds - pdV \\ T\frac{ds}{dt} &= V\frac{d\rho}{dt} + (\rho + p)\frac{dV}{dt} \end{aligned}$$

By substituting the continuity equation $\frac{d\rho}{dt} = -3H(\rho + p)$ into the above equation, we can get:

$$\begin{aligned} T\frac{ds}{dt} &= V(-3H(\rho + p)) + 3HV(\rho + p) \\ &= 0 \end{aligned}$$

Thus:

$$\frac{ds}{dt} = 0$$

which match our intuition that the entropy is conserved in the universe.

And by knowing the relationship between energy density and entropy density, we can derive the effective number of relativistic degrees of freedom for entropy density $g_{*s}(T)$ in the early universe.

Theory

For a relativistic particle species in thermal equilibrium, we have:

$$\rho = \frac{\pi^2}{30} g_*(T) T^4,$$

and using:

$$s = \frac{\rho + p}{T},$$

we get:

$$s = \frac{2\pi^2}{45} g_{*s}(T) T^3,$$

where $g_{*s}(T)$ is the effective number of relativistic degrees of freedom for entropy density, which is defined as:

$$g_{*s}(T) = \sum_{\text{bosons}} g_i + \frac{7}{8} \sum_{\text{fermions}} g_i$$

From the above equation, we can get the entropy conservation in another way:

FOUNDATION

Assume all particles remains relativistic and not decoupled when universe evolves, we have $g_{*s}(T) = \text{constant}$, thus we can get:

$$\rho \propto T^4,$$

Recall:

$$\rho \propto a^{-4},$$

we can get:

$$T \propto a^{-1},$$

At the same time, we also know that:

$$s \propto T^3,$$

Therefore:

$$sV \propto sa^3 = \frac{2\pi^2}{45} g_{*s}(T) T^3 a^3 = \text{constant}$$

5.4.3 g_* and g_{*s} in the early universe

FOUNDATION

In the early universe, if we assume all standard model particles are relativistic and in thermal equilibrium, we can calculate the effective number of relativistic degrees of freedom for energy density $g_*(T)$ and for entropy density $g_{*s}(T)$:

(1) For bosons:

- Photon: $g_\gamma = 2$
  (2 polarization states)
- Gluons: $g_g = 2 \times 8 = 16$
  (2 polarization states $\times$ 8 color states)
- W$\pm$ bosons: $g_W = 3 \times 2 = 6$
  (3 spin states $\times$ 2 charge states)
- Z boson: $g_Z = 3$
  (3 spin states)
- Higgs boson: $g_H = 1$
  (1 spin state)

(2) For fermions:

- leptons ((e, μ, τ), (e^+, μ^+, τ^+)): $g_l = 2 \times 3 \times 2 = 12$
  (2 spin states $\times$ 3 generations $\times$ 2 charge states)
- Neutrinos (ν_e, ν_μ, ν_τ): $g_\nu = 2 \times 3 = 6$
  (2 spin states $\times$ 3 generations, but neutrinos anti-particles are itself, so no charge states)
- Quarks ((u, d, c, s, t, b), ($\bar{u}$, $\bar{d}$, $\bar{c}$, $\bar{s}$, $\bar{t}$, $\bar{b}$)): $g_q = 2 \times 3 \times 6 \times 2 = 72$
  (2 spin states $\times$ 3 color states $\times$ 6 flavors $\times$ 2 charge states)

By summing over all relativistic species, we can get:

$$g_b = 2 + 16 + 6 + 3 + 1 = 28,$$
$$g_f = 12 + 6 + 72 = 90$$

And thus we can get:

$$g_*(T) = g_{*s}(T) = g_b + \frac{7}{8} g_f = 28 + \frac{7}{8} \times 90 = 106.75$$

THEORY

In early universe, by only considering particles in standard model, and assuming all particles are relativistic and not decoupled to other particles, we can get:

$$g_*(T) = g_{*s}(T) = 106.75$$

Important Note

The equivalence of $g_*(T) = g_{*s}(T)$ is only valid in the early universe that particles are not decoupled to other particles.
Again, if we consider the contribution of decoupled particles with different temperature, the equation will be:

$$\begin{cases} g_*(T) = \sum_{\text{bosons}} g_i \left(\frac{T_i}{T}\right)^4 + \frac{7}{8} \sum_{\text{fermions}} g_i \left(\frac{T_i}{T}\right)^4 \\ g_{*s}(T) = \sum_{\text{bosons}} g_i \left(\frac{T_i}{T}\right)^3 + \frac{7}{8} \sum_{\text{fermions}} g_i \left(\frac{T_i}{T}\right)^3 \end{cases}$$

and thus $g_*(T)$ and $g_{*s}(T)$ will no longer be equal, as they have different power of temperature ratio.

Question Example 5.1: Higgs Mechanism and g_*

We know that in Higgs mechanism, DoF of Higgs boson and W and Z bosons will changed, but our calculation of g_* in early universe is still valid, why?

Solution.
The answer is that both of them are still boson before and after Higgs mechanism, thus they will contribute to g_* with the same weight, and their total DoF is still remain unchanged, thus the value of g_* will not change.

Question Example 5.2: Current Entropy Density of the Universe

In the universe nowadays, the entropy density is dominated by photons bath due to its huge amount compared to other particles.
What is the current entropy density of the universe?

Solution.
Given the temperature of CMB is $T_{\text{CMB}} \approx 2.725\,\text{K} \approx 2.348\,\text{eV}$, only photons and neutrinos are still relativistic, thus we only need to consider the contribution of photons and neutrinos to the entropy density of the universe.

Degeneracy factor of photon:

$$g_\gamma = 2$$

and degeneracy factor of neutrinos:

$$g_\nu = 2 \times 3 = 6$$

Note due to e^+e^- annihilation only heating the protons but not neutrinos, the temperature of neutrinos is lower than that of photons by a factor of $(4/11)^{1/3}$,

by considering this, we can get:

$$g_{*s}(T_{\text{CMB}}) = g_\gamma + \frac{7}{8} g_\nu \left(\frac{T_\nu}{T_{\text{CMB}}}\right)^3 = 2 + \frac{7}{8} \times 6 \times \left(\frac{4}{11}\right) = 3.909$$

thus we can calculate the current entropy density of the universe as:

$$s_0 = \frac{2\pi^2}{45} g_{*s}(T_{\text{CMB}}) T_{\text{CMB}}^3 \approx 2.889 \times 10^9 \, \text{m}^{-3}$$

CHAPTER 6

Big Bang Nucleosynthesis (Lecture 13-14)

CHAPTER 7

Cosmic Microwave Background (Lecture 15-19)

CHAPTER 8

Dark Matter (Lecture 20-22)

CHAPTER 9

Inflation (Lecture 23-24)

CHAPTER A

Declaration of AI Assistance

A.1 Use of Artificial Intelligence in This Document

This course guide was developed with assistance from artificial intelligence (AI) in the following capacities:

- **Template Design:** AI provided guidance on structuring the document layout, choosing appropriate color schemes.
- **LaTeX Code Generation:** AI assisted in writing and refactoring LaTeX code, including package configurations, box styling (tcolorbox), TikZ decorations, and custom command definitions.
- **LaTeX Structure Refactoring:** AI helped reorganize and optimize the document structure for maintainability, readability, and professional presentation.
- **Mathematical Formula Formatting:** AI provided assistance in formatting and presenting mathematical equations in clear, readable notation aligned with conventions.

Author Verification: All content, including scientific accuracy, theoretical derivations, example problems, and pedagogical choices, has been carefully reviewed and verified by the author. The AI assistance was limited to technical formatting and structural optimization, not to the generation or validation of course content itself.

CHAPTER B

Mathematical Tools & Derivations

B.1 Useful Identities

Trigonometric

$$\sin^2\theta + \cos^2\theta = 1, \quad \tan\theta = \sin\theta/\cos\theta, \quad \sin(A \pm B) = \sin A\cos B \pm \cos A \sin B.$$

Hyperbolic

$$\cosh^2 x - \sinh^2 x = 1, \quad \sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y.$$

Common Integrals

$$\int x^n\,dx = \tfrac{x^{n+1}}{n+1} + C \quad (n \neq -1), \qquad \int e^{ax}\,dx = \tfrac{1}{a}e^{ax} + C,$$

$$\int \frac{dx}{x} = \ln|x| + C, \qquad \int \sin x\,dx = -\cos x + C.$$

Taylor Expansions

$$f(x) = f(a) + f'(a)(x-a) + \tfrac{f''(a)}{2!}(x-a)^2 + \cdots$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \tfrac{x^3}{3!} + \tfrac{x^5}{5!} - \cdots$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \tfrac{x^2}{2!} + \tfrac{x^4}{4!} - \cdots$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \tfrac{x^2}{2!} + \cdots$$

$$\sqrt{1+x} = \sum_{n=0}^{\infty} \binom{1/2}{n} x^n = 1 + \tfrac{1}{2}x - \tfrac{1}{8}x^2 + \cdots$$

B.2 Differential Geometry Primer

The **metric tensor** $g_{\mu\nu}$ encodes the geometry of spacetime: $ds^2 = g_{\mu\nu}\, dx^\mu\, dx^\nu$. The **Christoffel symbols** (not tensors) are defined as

$$\Gamma^\lambda_{\mu\nu} = \tfrac{1}{2} g^{\lambda\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}). \tag{B.1}$$

The **Ricci tensor** is the contraction $R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu}$ of the Riemann curvature tensor, and the Ricci scalar is $R = g^{\mu\nu} R_{\mu\nu}$. These enter the Einstein field equations as $G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = (8\pi G/c^4) T_{\mu\nu}$.

B.3 Relativistic Limits

The energy-momentum relation is $E^2 = p^2c^2 + m^2c^4$. So in a relativistic limit, we can have:

$$\{E, m, \rho\} \ll \{p, T\}$$

which any one of the values in left-hand set is much smaller than any one of the values in the right-hand set.

CHAPTER C

Constants & Conversion Factors

C.1 Physical Constants

Quantity	Symbol	Value (SI)
Speed of light	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{m^{3}\,kg^{-1}\,s^{-2}}$
Hubble constant today	H_0	$67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$
Cosmological constant	Λ	$1.11 \times 10^{-52}\,\mathrm{m^{-2}}$
Cosmological constant	Λ	$1.19 \times 10^{-35}\,\mathrm{s^{-2}}$
Planck constant	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s}$
Boltzmann constant	k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}}$

C.2 Conversion Factors

Conversion	Value
1 astronomical unit (AU)	$1.496 \times 10^{11}\,\mathrm{m}$
1 light-year (ly)	$9.461 \times 10^{15}\,\mathrm{m}$
1 parsec (pc)	$3.086 \times 10^{16}\,\mathrm{m}$
1 parsec (pc)	$3.262\,\mathrm{ly}$
1 year (yr)	$3.156 \times 10^{7}\,\mathrm{s}$
1 eV	$1.602 \times 10^{-19}\,\mathrm{J}$
1 electron mass (m_e)	$0.511\,\mathrm{MeV}/c^2$
1 electron mass (m_e)	$9.109 \times 10^{-31}\,\mathrm{kg}$
1 proton mass (m_p)	$938.3\,\mathrm{MeV}/c^2$
1 proton mass (m_p)	$1.673 \times 10^{-27}\,\mathrm{kg}$
1 neutron mass (m_n)	$939.6\,\mathrm{MeV}/c^2$
1 neutron mass (m_n)	$1.675 \times 10^{-27}\,\mathrm{kg}$
1 Solar mass ($M_\odot$)	$1.989 \times 10^{30}\,\mathrm{kg}$
1 solar radius ($R_\odot$)	$6.957 \times 10^{8}\,\mathrm{m}$
1 Earth mass ($M_\oplus$)	$5.972 \times 10^{24}\,\mathrm{kg}$
1 Earth radius ($R_\oplus$)	$6.371 \times 10^{6}\,\mathrm{m}$
CMB temperature (T_{CMB})	$2.725\,\mathrm{K}$

C.3 SI Unit Prefixes

Prefix	Name	Factor
10^{-9}	nano (n)	10^{-9}
10^{-6}	micro (μ)	10^{-6}
10^{-3}	milli (m)	10^{-3}
10^{3}	kilo (k)	10^{3}
10^{6}	mega (M)	10^{6}
10^{9}	giga (G)	10^{9}
10^{12}	tera (T)	10^{12}

REFERENCES

[1] Moshe Carmeli. "Value of the cosmological constant: Theory versus experiment". In: *AIP Conference Proceedings*. Vol. 586. AIP, 2001, pp. 316–318. DOI: 10.1063/1.1419571. URL: http://dx.doi.org/10.1063/1.1419571.

[2] *PHYS4071 Big Bang Cosmology & Inflation Lecture Notes*. 2026.